COURSE INFORMATION

School/Faculty:	**Faculty of Computing**		
Program name:	**Bachelor in Sofwtare Engineering with Honours**		
Course code:	**SCSP3213**	**Academic Session/Semester:**	**2025/2026-2**
Course name:	**Business Intelligence**	**Pre/co requisite** (*course name and code, if applicable*):	
Credit hours:	**3**		

Course synopsis	This course introduces students to the concepts, architectures, and technologies used in Business Intelligence (BI) to transform organizational data into meaningful insights for decision-making. It is an intermediate-level course that integrates data engineering principles with analytical and visualization techniques used in modern BI environments. The course covers key BI components including operational data sources, ETL processes, data warehousing platforms, and big data technologies. Students explore data analytics applications, predictive analytics techniques, and front-end BI tools for reporting and visual discovery. The course also emphasizes dashboard design, visual analytics, and storytelling using tools such as Alteryx, Tableau, and Power BI. The course is delivered through lectures, laboratory sessions, active learning activities, industry talks and project-based learning. Upon completion, students will be able to analyze complex datasets, design BI workflows and dashboards, and communicate actionable insights to support data-driven decision-making in organizations.
Course coordinator (*if applicable*)	Assoc. Prof. Dr. Noorminshah Bt A.Iahad□

Course lecturer(s)	**Name**	**Office**	**Contact no.**	**E-mail** (@utm.my)
	Assoc. Prof. Dr. Noorminshah Bt A.Iahad	N28-402-06	013-3581955	minshah
	Dr. Izyan Izzati Bt Kamsani	N28a-02-40-01	013-7493793	izyan
	Dr. Shamini A/P Raja Kumaran	N28-403-22	013-5213941	shamini.rk

Mapping of the Course Learning Outcomes (CLO) to the Programme Learning Outcomes (PLO), Teaching & Learning (T&L) methods and Assessment methods:

No.	CLO	PLO (MQF Cluster Code)	Taxonomies and generic skills	T&L methods	Assessment methods
CLO1	Explain relevant concepts, principles, and theories in Data Engineering and Business Intelligence using systems thinking to address given problems.	PLO1 (C1)	C2 SC1	Lecture, Active learning, Industry talk	Asg 1 (5%), Q (10%), Asg 2 (5%) F (5%)
CLO2	Demonstrate integrated problem-solving skills for complex data challenges and show competency in using assigned Business Intelligence software.	PLO3 (C3A)	P4 SC3	Lab Work,Project-based learning	Lab (5%) PR (10%), E (10%) F (15%)
CLO3	Construct an effective data analysis workflow and visualizations by applying systems thinking to generate actionable insights in the form of tables, charts, or dashboards using the assigned BI software.	PLO6 (C3D)	C6 A(IT2)	Lab Work, Project based learning	Lab (10%), PR (10%) F (10%)

Refer ***Taxonomies of Learning and ****UTM's Graduate Attributes for UG and Generic Skills for PG, where applicable for measurement of outcomes achievement
*****L-Lab Assignment ; Q – Quiz; E - Exam , Asg – Assignment; PR – Project; F – Final Exam

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<MQF Cluster Code>
C1 = Knowledge & Understanding, C2 = Cognitive Skills, C3A = Practical Skills, C3B = Interpersonal Skills, C3C= Communication Skills, C3D = Digital Skills, C3E = Numercy Skills, C3F = Leadership, Autonomy & Responsibility, C4A = Personal Skills, C4B = Enterpreneurial Skills, C5 = Ethics & Professionalism
<Taxonomies>
Cognitive (C) - C1-C6, Psychomotor (P) - P1-P7, Affective (A) - A1-A5
<Graduate Attributes>
SC1, SC2, SC3 = Scholars, SI1, SI2, SI3, SI4 = Social Intelligence, IT1, IT2, IT3 = Innovative Talent, CC1, CC2, CC3, CC4 = Constructive Citizen, ES1, ES2, ES3, ES4, ES5 = Enterprising Skills

Details on Innovative T&L practices:

No.	Type	Implementation
1	Active learning	In-class activities, including Think–Pair–Share, Jigsaw, Gallery Walk, and Predict–Observe–Explain, are implemented to enhance student engagement, collaboration, and understanding of business intelligence concepts. These activities provide opportunities for students to analyze real-world datasets, apply BI tools and collaboratively design dashboards and reports.
2	Project-based learning	Implemented through a real industry case study with tasks provided step by step throughout the semester. Students work in groups to create a data preprocessing workflow and design a dashboard using BI software. This approach helps students develop analytical, problem-solving, and communication skills while applying BI techniques to real-world business problems.
3	Industry Talk	Industry talk (or Industry Guest Lecture) is an approach of work-based learning. A sharing session by industry professionals provides students with exposure to current Business Intelligence technologies, real-world practices, and implementation strategies used in the industry.
4	Lab Work	Hands-on BI practical sessions to performing data cleaning, modeling and creating dashboards.

Transferable skills (*generic skills learning in course of study which can be useful and utilised in other settings*) **:**
Apply Business Intelligence technologies for effective data analysis in problem-solving

Student Learning Time (SLT) / Effective Learning Time (ELT) details:

Week/ Meeting	Course Content Outline and Subtopics	CLO*	Learning and Teaching Activities									TOTAL ELT
			Face-to-Face (F2F)								Non F2F Independent Learning	
			Physical				Online (Synchronous)				Online (Asynchronous)	Others
			L	T	P	O	L	T	P	O		
Week 1	Chapter 1: Introduction to Business Intelligence 1.1 Introduction to Business Intelligence (BI) 1.2 Concept, Effect of BI in Business 1.3 The Importance of BI Tools 1.4 Business Intelligence Governance	CLO1					3					2	5
Week 2	Chapter 2: Components of Business Intelligence Architecture 2.1 Operational and Source System 2.2 Data Transfer: From Operational to Data Warehouse 2.3 The Data Warehouse & Technology Platform 2.4 Big Data Technologies	CLO1					3					2	5
Week 3	Chapter 3: Business Intelligence Front-End 3.1 Self-Service BI 3.2 Business Query and Reporting 3.3 Visual Data Discovery 3.4 Dashboard 3.5 Scorecard 3.6 Online Analytical Processing (OLAP) 3.7 Analytic Applications 3.8 Big Data Analytics	CLO1	3									2	5
Week 4	Industry Talk	CL01	3									2	5
Week 5	Chapter 4: Data Analytic Applications for Business 4.1 Access & Writing Data 4.2 Joining and Data Prep 4.3 Data Formula and Calculation Lab Session 1	CLO2	1		2						1	2	6
Week 6	Chapter 4: Data Analytic Applications for Business (cont'd) 4.4 Data Investigation Lab Session 2	CLO2	1		2						1	2	6
Week 7	Professional Cert (Alteryx Designer – Core Level)	CLO2							3			5	8
Week 8	**MID SEMESTER BREAK**												
Week 9	Chapter 4: Data Analytic Applications for Business (cont'd) 4.5 Predictive Analytics - Regression, Classification Lab Session 3	CLO2			3						1	2	6
Week 10	Chapter 4: Data Analytic Applications for Business (cont'd) 4.6 Predictive Analytics - Clustering Lab Session 4	CLO2			3						1	2	6
Week 11	Industry Talk	CLO1	3									1	4
Week 12	Chapter 5: Building Visual Models 5.1 Core Concepts of Visual Modeling 5.1.1 Importance of Visual Models in Data Analysis 5.1.2 Types of Visual Representations (Charts, Graphs, Maps) 5.1.3 Data Visualization Best Practices 5.1.4 Understanding User Experience in Visual Analytics	CLO3	1		2						1	2	6
Week 13	Chapter 5: Building Visual Models 5.2 Dashboard Design & Storytelling in Tableau 5.2.1 Introduction to Tableau's Interface 5.2.2 Connecting and Preparing Data 5.2.3 Creating Interactive Visualizations 5.2.4 Applying Filters and Parameters 5.2.5 Storytelling Techniques with Dashboards	CLO3	1		2						1	2	6
Week 14	Chapter 5: Building Visual Models 5.3 Dashboard Design & Storytelling in Power BI 5.3.1 Overview of Power BI Workspace 5.3.2 Data Import and Transformation 5.3.3 Designing Effective Visuals and Reports 5.3.4 Using Slicers, Filters, and DAX for Interactivity 5.3.5 Best Practices for Storytelling with Power BI Dashboards Lab Session 5	CLO3	1		2						1	2	6
Week 15	5.3 Publish Workbook	CLO3	3										3
											SUB-TOTAL SLT :		**77**

	Continous Assessment	%	Face-to-Face (F2F)		NF2F Independent Learning for Assessment		TOTAL ELT
			Physical	Online (Synchronous)	Online (Asynchronous)	Others	
1	Quiz	10	1			2	3
2	Assignment 1	5				1	1
3	Assignment 2 (Industry Talk)	5				1	1
4	Lab Assignment (x3) □	15				9	9
5	Exam (Pro-Cert)	10			3	3	6
6	Project	20			4	8	12
7	Peer Assessment	5			1		1

SUB-TOTAL SLT :	**33**

	Summative Assessment	%	Face-to-Face (F2F)		NF2F Independent Learning for Assessment		TOTAL ELT
			Physical	Online (Synchronous)	Online (Asynchronous)	Others	
1	Final Exam	30	3		2	5	10
					SUB-TOTAL SLT :		**10**
					SLT for Assessment:		**43**
					GRAND TOTAL SLT:		**120**

A	% ELT for F2F Physical Component:	30.83
B	% ELT for Online & Independent Learning Component :	69.17
C	%ELT for Online Component:	21.67
D	% ELT for All Practical Component:	15.83
D1	% ELT for F2F Physical Practical Component:	13.33
D2	% ELT for F2F Online Practical Component:	2.50
Please tick (/) if this course is Industrial Training / Clinical Placement / Practicum using 50% of Effective Learning Time (ELT)		

Identify special requirement or resources to deliver the course (*e.g.,software,nursery, computer lab, simulation room etc*)

Computer lab; Software (Alteryx Designer, Tableau Desktop, Power BI)

References

Main References:

Howson, C. (2023). Successful business intelligence (2nd ed.). McGraw Hill.
Burkhow, J. (2023). Alteryx Designer: The definitive guide. O'Reilly.
Mamta, M., & Raheja, N. G. (2024). Data visualization and storytelling with Tableau. CRC Press.

Additional References:

Charles, V., Garg, P., Gupta, N., & Agarwal, M. (2023). Data analytics and business intelligence. CRC Press.

Online:

http://elearning.utm.my

Other additional information (*if applicable*)

Academic honesty and plagiarism:

Assignments are individual tasks and NOT group activities (UNLESS EXPLICITLY INDICATED AS GROUP ACTIVITIES)
Copying of work (texts, simulation results etc.) from other students/groups or from other sources is not allowed. Brief quotations are allowed and then only if indicated as such. Existing texts should be reformulated with your own words used to explain what you have read. It is not acceptable to retype existing texts and just acknowledge the source as a reference. Be warned: students who submit copied work will obtain a mark of zero for the assignment and disciplinary steps may be taken by the Faculty. It is also unacceptable to do somebody else's work, to lend your work to them or to make your work available to them to copy.

Other additional information (*if applicable*)

All students must adhere to the following to pass a computing course:
i. Pass both Continuous Assessment AND Final Assessment.
ii. Get at least minimum pass marks following UTM regulation (40% or D+).
The passing mark for:
- Continuous Assessment is 30 % of continuous marks.
- Final Assessment is 20% of the final marks.
If a student does not achieve the passing mark for Continuous Assessment OR the passing mark for the Final Assessment, then the student's overall final mark will be set below the UTM passing mark.-

Disclaimer:

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ELT = (Theory + Industrial Guidance + Assessment) x 80%

Total of credit for LI/Practical = ELT/40 Notional Hours

Note: For ODL Programme : Courses with mandatory practical requirement imposed by programme standards or any related standards can be exempted from complying to the minimum 80% ODL delivery rule in the SLT.

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Signature: *noorminshah*	**Signature:**
Date: 09/03/2026	**Date:** 09/03/2026